

BHAGIRADH KANTHETI

+1 9707100106 | bhagiradh.1994@gmail.com
7010 Buckhorn Cir, Colorado Springs, CO 80919

*AI/ML - Engineer and PhD candidate
with 8 years of experience in designing, building, implementing and deploying advanced AI models*

PROFESSIONAL SUMMARY

AI/ML - Engineer and PhD candidate with 8 years of experience in developing and implementing advanced AI models. Expertise in developing machine learning and deep learning models, optimizing algorithms, fine tuning LLMs and deploying scalable solution on cloud platforms. Proven ability to build a project from concept to production, ensuring performance, scalability and efficiency in real world applications. Experience in delivering products or solutions that utilized Machine Learning, Natural Language Processing or other forms of AI. Experienced in containerization, orchestration and deployment process with strong foundation in Python based frameworks and cloud technologies.

CORE COMPETENCES

- **AI:** Deep Learning, Transformers, Large Language Models (LLM's), Restricted Boltzmann Machines (RBM), RNN, CNN, LSTMs, Linear Regression, XGBoost, LightGBM, Naive Bayes, SVM, Decision Forests, Natural Language Processing (NLP), Prompt Engineering, Retrieval Augmented Generation (RAG), Agentic AI, Time series Forecasting, Supervised and Unsupervised learning, Computer Vision (CV), Fine Tuning, Creating Pipelines, Experimental Design, Optimizations, Hyperparameter Tuning, Fine Tuning, Model Evaluation, Private Inference, Research Methodology.
- **Data Analysis and Processing:** Data cleaning and preprocessing, Statistical analysis, Feature Engineering, Stationarity Testing, Outlier Detection, Autocorrelation analysis, and Data Compression.
- **Programming Language:** Python, SQL, MATLAB.
- **IDE s/ Text Editors:** Jupiter, PyCharm, Visual Studio, Excel, Spyder, RStudio, MATLAB, Notepad++, Cube IDE
- **Frame Works and Libraries:** PyTorch, Keras, FastAPI, Scikit-learn, Hugging Face Transformers, Statsmodels, SciPy optimize, Pandas, NumPy, Bio python, NLTK, SpaCy Matplotlib, Seaborn.
- **Database:** SQL, PostgreSQL
- **Cloud and DevOps:** Docker, Amazon Web Services (AWS), Google Cloud Platform (GCP), Azure.
- **Web Development:** Flask, HTML, CSS
- **Specialized Tools:** FAISS, LangChain, CrypTen, Keil and ARM Forge.

PROFESSIONAL EXPERIENCE, KEY SKILLS, AND ACCOMPLISHMENTS

2021-01-2024-12

AI/ML Engineer/Researcher

Algemetric INC, Colorado Springs

- **NLP and Text-to-SQL Translation for Database Interactions**
Key Skills: Natural language processing (NLP), T5 (LLM), dynamic schema handling
 - Architected and implemented Text-to-SQL generator using T5 transformer model with PyTorch and Hugging Face
 - Developed dynamic schema handling system supporting multiple database tables using SQLAlchemy
 - Created comprehensive prompt engineering pipeline with custom templates and validation
 - Implemented containerization using Docker and Docker Compose with multi-stage builds

- Established REST API endpoints using FastAPI with OpenAPI documentation
- Integrated PostgreSQL for query validation and performance testing
- Implemented automated model retraining pipeline with data drift detection
- Deployed system using Kubernetes for scalability and management with automated GitLab CI/CD pipeline
- **Slot machine compression for Multi FASTA Files**
Key Skills: DNA Sequencing, Multi FASTA Files, Genomic Data Compression
 - Worked on optimizing genomic data compression for multi-FASTA files used in DNA sequencing.
 - Achieved up to 80.96% compression of DNA sequences, ensuring efficient data retrieval without the need for full decompression.
 - Utilized bioinformatics techniques to enhance random access and storage efficiency for large-scale genomic data.
- **Interactive generative AI for Financial Forecasting as a feature**
Key Skills: Retriever-Augmented Generation (RAG), OpenAI API (LLM), Natural Language Processing (NLP), FAISS, LangChain, Python.
 - Designed and implemented RAG system architecture integrating OpenAI's GPT-4 API
 - Built vectorized document retrieval system using FAISS for high-accuracy similarity search
 - Developed custom LangChain components for context-aware retrieval
 - Implemented logging and experiment tracking using Weights & Biases
 - Implemented DVC for model and data versioning with Git integration
 - Deployed on AWS using ECS/ECR with auto-scaling capabilities
 - Created automated model monitoring and retraining pipeline
 - Improved trend analysis accuracy by 15% over baseline models
- **Advancing Transformers for Time Series Forecasting and Private Inference**
Key Skills: Secure multi-party computation (MPC) with CrypTen, Transformers, PyTorch, cloud computing
 - Developed novel transformer architecture using PyTorch and CrypTen for secure computation
 - Implemented distributed Inference using PyTorch across multiple GPU instances
 - Created secure inference pipeline using Multiparty Computation (MPC) techniques
 - Deployed across AWS EC2 (p3.2xlarge) and GCP (Tesla V100) instances
 - Built automated CI/CD pipeline using AWS and Docker
 - Created custom performance metrics and visualization tools
 - Reduced inference time by 15% and communication overhead by 60%
 - Published research findings in peer-reviewed conference
- **Optimizing Data Pipelines and Forecasting Accuracy**
Key Skills: Data preprocessing, cloud-based pipelines, statistical tests, model experimentation
 - Designed comprehensive ETL pipeline using Apache Airflow and Apache Kafka
 - Implemented advanced time series analysis using statsmodels (ADF, ACF, PACF)
 - Created automated feature engineering pipeline using scikit-learn
 - Built model selection framework using GridSearchCV and RandomizedSearchCV
 - Implemented data validation using Great Expectations
 - Developed A/B testing framework with statistical hypothesis testing
 - Created automated performance reporting using custom dashboards
 - Deployed using Kubernetes for orchestration and scaling
 - Improved forecast accuracy by 10% on weather datasets
- **AI-Driven Revenue Optimization and ETL Pipeline Development**
Key Skills: Python-based constraint optimization, profitability analysis, ETL pipeline creation
 - Designed an AI-driven revenue optimization tool leveraging Python, Scikit-learn, and Pandas for linear regression and constraint optimization, enhancing profitability by identifying optimal revenue streams.

- Built an ETL pipeline with Pandas, ensuring efficient data loading, cleaning, and processing from six different sources.
- Integrated technologies like NumPy and Matplotlib for data validation and visualization to maintain data quality and accessibility.

2019-09 - 2020-12

Electrical Engineer

Symetrix Corporation, Colorado Springs

- **Optimizing Neural Network Inference on ARM Cortex-M Microcontroller**
Key Skills: Embedded systems, neural network deployment, ARM Cortex-M, Keil tool, Python/Embedded C
 - Deployed a Restricted Boltzmann Machine (RBM) neural network inference onto an ARM Cortex-M microcontroller using Keil development tools.
 - Converted floating-point operations to integer-based inference, significantly improving processing efficiency on embedded systems.
 - Conducted time analysis tests, achieving 42x faster performance and 62x lower energy consumption.
- **Conducting Research on Quantum Computing and Compression**
Key Skills: Quantum gates, geometric algebra, quantum computing research
 - Conducted in-depth research on quantum gates and circuits, exploring Geometric Algebra (Clifford Algebra) for advanced quantum computing techniques.
 - Investigated quantum compression methodologies for potential applications in AI and data science.

2018-06 - 2019-08

Software Development Engineer

Machine Learning & AI

Symetrix Corporation, Colorado Springs

- **Developing and Optimizing Machine Learning Algorithms**
Key Skills: Machine learning algorithm development, data preprocessing, Python
 - Programmed a proprietary machine learning algorithm in Python, leveraging PCA, Pearson correlation, and Normal Distribution techniques for advanced data preprocessing.
 - Applied nonparametric statistics for parameter estimation during training, optimizing classification accuracy and computational efficiency.
- **Training and Classification with Restricted Boltzmann Machine (RBM)**
Key Skills: RBM training, dataset classification, model optimization
 - Trained Restricted Boltzmann Machines (RBM) using various datasets, including MNIST, ORL Face Database, CIFAR-10, and Google Speech Commands.
 - Developed an integer-based RBM using quantization and scaling techniques, achieving significant performance optimization and accuracy improvements over industry benchmarks.

2017-07 - 2018-05

Image Processing Assistant (Intern)

Symetrix Corporation, Colorado Springs

- **Developing Face Recognition Algorithm Using Neural Networks**
Key Skills: Image processing, face recognition, Bezier curves, edge detection
 - Created a face recognition algorithm by integrating Bezier curve and edge detection techniques with Neural Networks.
 - Reduced face representation to 100 coordinates, significantly enhancing processing efficiency and maintaining high accuracy

EDUCATION

Ph.D. Electrical Engineering University of Colorado at Colorado Springs

January 2021-Present

Coursework: **Deep Learning**, Advanced Geometric Algebra for Quantum Computing (Independent study 1), **Neuroscience-informed Modeling and Simulation for new AI Platform** (Independent study 2), Signals and Systems.

M.S Electrical Engineering University of Colorado at Colorado Springs

Aug 2015-May 2017

Coursework: VLSI Circuit Design-I, Analog IC Design, Electromagnetic Effects in IC Design, Advanced Digital Design, RF circuit design, Computer Architecture & Design, Microwave Circuit Design, Computer/Network Security, Independent Study on IOT, Electronic Properties of Materials.

B. Tech. Electronics and Instrumentation. VNR Vignana Jyothi College Of E & T, India

2011-2015

ADDITIONAL ACCOMPLISHMENTS

Bhagiradh Kantheti., Dr. Carlos Paz De Araujo., (2024). “Confidential Trends: Assessing the Feasibility of Transformers in Private Time Series Forecasting” was peer-reviewed and accepted for the 7th International AICCC Conference.

Oliveira, H., Delavignette, G., Harmon, L., **Kantheti, B.**, (2024). "Slot Machine: A Compression Transform for MultiFASTA Files". The manuscript was peer-reviewed and accepted for the 15th International Conference on ICT Convergence (ICTC).